A Descriptive and Inferential Analysis of AI Adoption, Work Models, and Compensation in the Software Industry

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***Abstract*—This report presents a statistical analysis of the software engineering workforce using the 2024 Stack Overflow Developer Survey (N=20,754). After an initial descriptive phase, we applied hypothesis testing (T-Test, ANOVA) and Multiple Linear Regression (OLS) to examine the factors affecting developer compensation. The analysis specifically focuses on the financial impact of Artificial Intelligence (AI) tools. Our results indicate that traditional factors, such as years of professional coding experience and formal education, are still the main predictors of salary. Interestingly, when controlling for seniority and education, the initial salary gap observed between AI users and non-users loses statistical significance ($p = 0.587$). This suggests that higher average salaries among non-users are related to their higher average experience levels rather than the absence of AI tools. Additionally, remote work models were found to correlate with higher median salaries compared to hybrid or in-person setups.**

I. Introduction

A. Overview

The software industry is continuously evolving, and the profile of the average developer is changing along with it. The 2024 Stack Overflow Developer Survey provides a large-scale dataset with responses from over 65,000 participants. This study uses a subset of this data to examine the demographics, tool preferences, and working environments of software developers.

B. Motivation

A computer science degree is not the only way to enter the industry today, as self-taught developers make up a noticeable portion of the workforce. Also, AI-assisted development tools are becoming more common in daily coding tasks.

This raises questions about adoption patterns. Do experienced developers use these tools as much as beginners? Furthermore, it is worth investigating if using AI tools has a measurable effect on financial compensation. This study looks at how education, experience, and work models relate to salary and AI adoption.

C. Research Questions

To guide this analysis, the following research questions (RQs) were established:

- **RQ1:** Is there a noticeable difference in the accumulated years of professional experience between developers with formal university education and those from alternative educational paths?
- **RQ2:** How does a developer's level of professional experience relate to their likelihood of adopting AI-based coding tools?
- **RQ3:** Is there a noticeable difference in annual compensation between developers who actively integrate AI into their workflows and those who do not?
- **RQ4:** How do different work environments (Remote, Hybrid, In-person) influence the distribution of developer salaries?
- **RQ5:** Which popular programming language communities exhibit the highest rates of AI tool adoption?

D. Contributions

The first part of this study relied on descriptive statistics. The main contribution of this second report is the addition of inferential analysis. By using linear regression and hypothesis testing, the study attempts to better isolate the variables affecting compensation. Specifically, it tests whether the adoption of AI tools has an independent effect on a developer's salary when controlling for established factors like seniority and formal education.

II. Method

A. Dataset

1) Overview: The dataset used in this study is derived from the May 2024 Stack Overflow Developer Survey. To enable quantitative analysis, we had to encode categorical and textual responses into numerical formats (using binary and label encoding where appropriate).

During the preprocessing phase, we removed any records missing values for our key variables. We also had to deal with extreme salary outliers. To handle this, we used a range-based approach, restricting the annual compensation to values between $10,000 and $500,000. While this range is somewhat arbitrary, we chose it to filter out unrealistic or highly noisy entries without destroying the natural shape of the income distribution.

After this cleaning process, we were left with 20,754 valid observations for the inferential analysis. It is worth noting, however, that because this dataset relies entirely on self-reported survey responses, there is an inherent risk of reporting bias.

2) Attributes: To answer our research questions, we narrowed our focus down to a few specific variables:

- **EdLevel_Numeric:** We created this simple binary flag to separate developers with a formal university degree (1) from those who are self-taught or followed alternative paths (0).
- **YearsCodePro:** A continuous variable that tracks how many years a respondent has been coding professionally.
- **AISelect_Numeric:** Another binary flag indicating whether the developer actively uses AI tools in their workflow (1) or not (0).
- **ConvertedCompYearly:** Our primary dependent variable, tracking annual compensation converted to USD.
- **DevType:** The respondent's primary professional role (e.g., Full-stack, Back-end).
- **LanguageHaveWorkedWith:** A list of programming languages the developer currently works with.
- **RemoteWork:** The current working environment, categorized as Remote, Hybrid, or In-person.

B. Method for Answering Research Questions

To move beyond basic observations, we relied on three statistical tests:

- **Independent Samples T-Test (Welch's):** We used this to see if the average salaries of AI users and non-users were significantly different from each other.
- **One-Way ANOVA:** This allowed us to compare mean salaries across the three different work environments.
- **Multiple Linear Regression (OLS):** This served as our core analytical tool. By controlling for multiple factors (education, experience, and work models) at the same time, the OLS model helped us isolate the true financial impact of adopting AI tools.

For all of these statistical tests, we set our significance threshold (α) at the standard 0.05 level.

III. Results

A. Descriptive Analysis

A set of descriptive statistical techniques was applied to explore the underlying structure of the dataset. Measures such as mean, median, variance, and standard deviation were computed, followed by visual analysis.

We first examined the distributions of professional experience and income. A comparison of experience across educational backgrounds (Fig. 1) showed that formal university education does not necessarily lead to a longer presence in the industry. The spread of experience is surprisingly similar for both university graduates and self-taught developers. Regarding compensation, the histogram and KDE plot (Fig. 2) reveal a strongly right-skewed distribution. This matches typical income patterns where

most individuals earn moderate salaries, with a long tail of high earners.

When looking at tool adoption, initial bar chart comparisons (Fig. 3) indicated that developers who avoid AI tools tend to have slightly higher average salaries. However, as we suspected early on, this difference should be interpreted cautiously; it is likely heavily influenced by seniority, since older, highly paid developers might simply be adopting AI at a slower rate. In terms of demographics, the dataset is largely dominated by Full-stack, Back-end, and Front-end developers (Fig. 4). Furthermore, AI adoption varies noticeably across different programming languages (Fig. 5), possibly due to how easily these tools integrate into specific ecosystems. Finally, work models show distinct salary variations (Fig. 6), with remote positions demonstrating noticeably higher median salaries compared to hybrid or in-person roles.

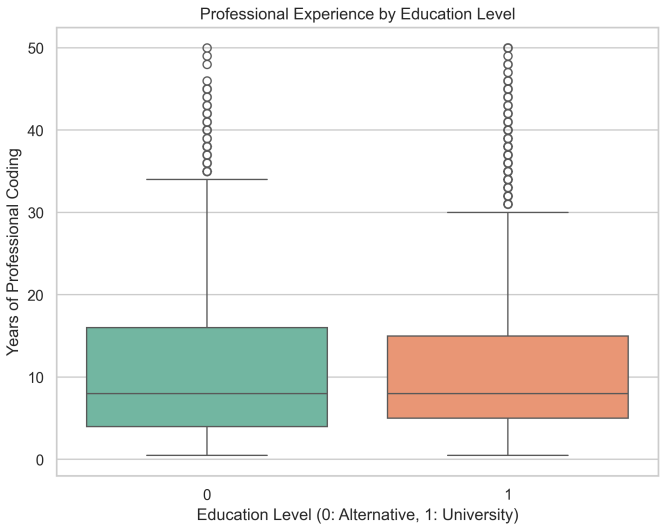

Fig. 1. Distribution of professional experience years according to educational backgrounds.

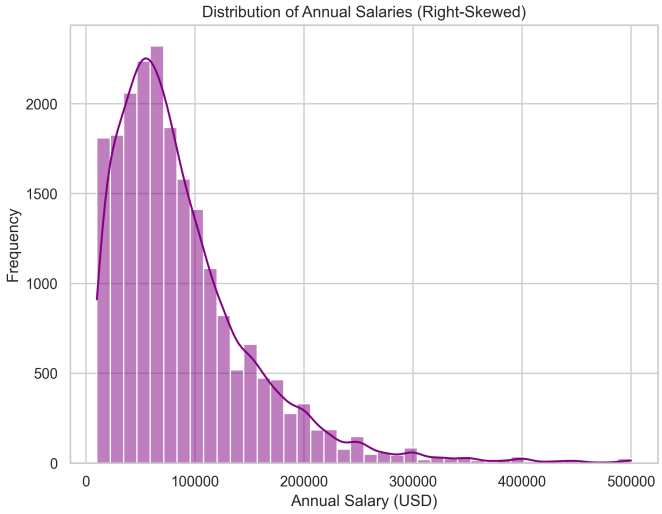

Fig. 2. Right-skewed distribution of annual salaries (USD).

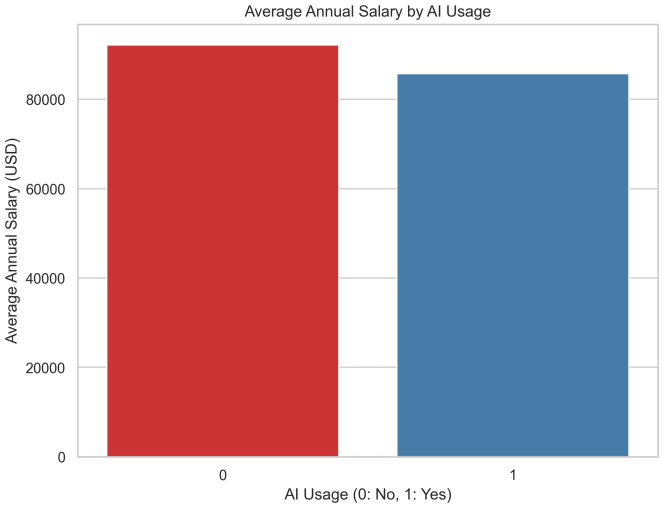

Fig. 3. Average annual salary comparison based on AI usage.

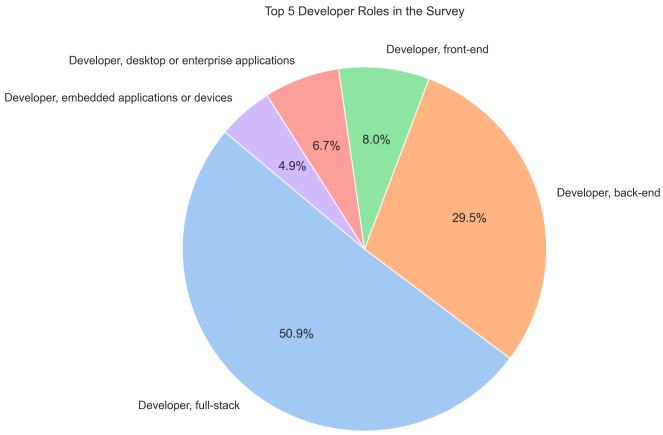

Fig. 4. Distribution of the top 5 developer roles.

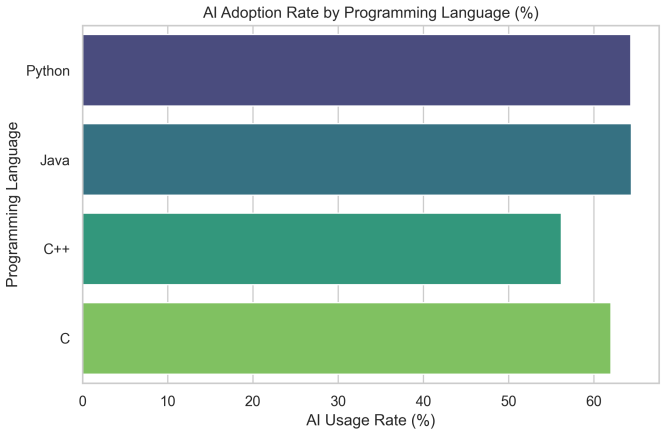

Fig. 5. AI adoption rates among popular programming languages.

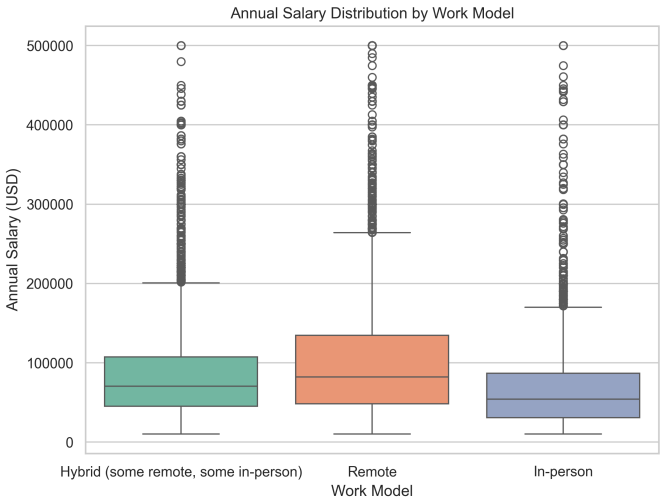

Fig. 6. Annual salary distribution compared by work models.

B. *Analysis for Answering Research Questions*

1. The Effect of Experience and Education (RQ1 & RQ2): When we ran the regression model, it became pretty clear that traditional background factors still heavily dictate compensation. For example, holding everything else constant, having a formal university degree is associated with an average base salary increase of \$13,640 ($p < 0.001$). But experience is the real driver here; every extra year of professional coding translates to an approximate increase of \$2,498 ($p < 0.001$). You can see this upward trend clearly in the scatterplot (Fig. 9).

2. Work Model Differences (RQ4): Our ANOVA test results ($p = 1.52 \times 10^{-123}$) confirmed that there is a massive, statistically significant difference in salaries across work environments. According to the OLS regression, fully remote developers earn about \$11,120 more on average compared to hybrid roles ($p < 0.001$). Conversely, strictly in-person roles earn approximately \$10,620 less ($p < 0.001$).

3. AI Usage and Compensation (RQ3): This is arguably the most intriguing part of the analysis. At first glance, a simple T-Test ($p = 8.33 \times 10^{-12}$) told us there was a significant salary difference between AI users and non-users. However, checking the correlation matrix (Fig. 7) revealed a negative correlation (-0.14) between experience and AI adoption.

Once we actually accounted for experience and education as control variables in our OLS regression, the coefficient for AI adoption plummeted to -\$473 with a p-value of 0.587. Since this is well above our 0.05 threshold, the effect of AI usage on salary is not statistically significant. Looking at the coefficient plot (Fig. 8), you can see the confidence interval for AI usage crossing right over the zero line. Ultimately, this confirms our earlier suspicion: the higher salaries we saw among non-AI users in the descriptive phase are just a byproduct of their seniority.

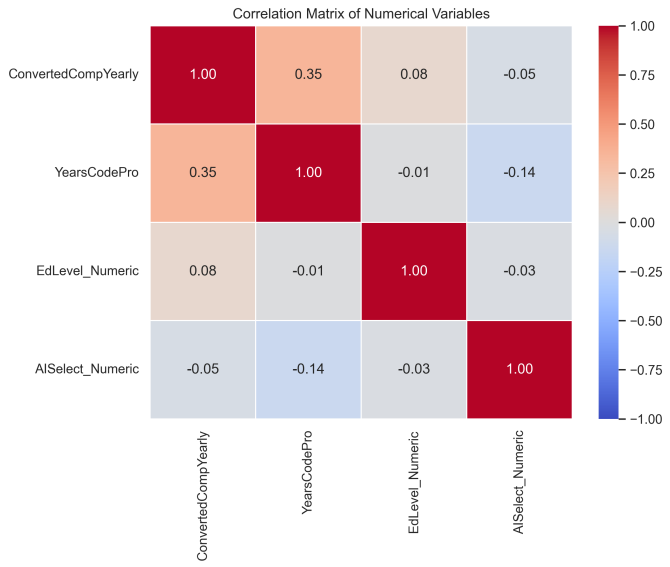

Fig. 7. Correlation Matrix highlighting relationships between numerical variables.

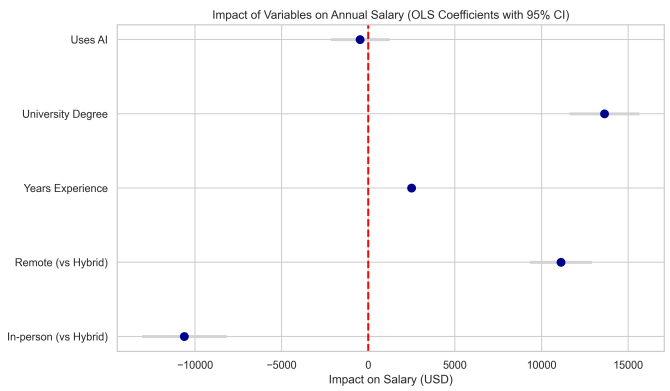

Fig. 8. Impact of variables on annual salary (OLS Coefficients with 95% CI).

Fig. 9. Linear Regression model demonstrating the positive effect of experience on salary.

IV. Conclusion

A. Summary and Future Works

Looking closely at the 2024 Stack Overflow Developer Survey, this study highlights that despite the rapid rise of new technologies, traditional metrics still largely rule the job market. Professional experience and a formal university education remain the most reliable predictors of a developer's income. We also saw that fully remote setups offer a clear financial edge over hybrid or office-based roles in this dataset.

The most interesting takeaway, however, relates to AI tools. While it initially looked like non-AI users were earning more, our regression analysis showed that this is actually an illusion caused by seniority. Older, more experienced developers simply earn higher salaries and happen to adopt AI tools a bit more slowly. When we control for that experience, the tools themselves don't currently provide any statistically significant salary bump.

Future Works: Admittedly, our current OLS model only explains about 14.3% of the variance in salaries ($R^2 = 0.143$). While this might seem low, it is actually quite expected when dealing with global salaries that fluctuate wildly based on local economies and costs of living. A logical next step for this research would be to split the dataset by geographic regions or country income levels. Factoring in those local economic conditions would likely produce a much tighter model and clear up some of the unobserved noise.

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